CBE 30235: Introduction to Nuclear Engineering Problem Set 3: Alpha Decay, Radon Transport, and Geochronology

Due: Friday, February 7, 2026 (via GradeScope at Midnight)

Problem 1: Counting Statistics and the Limit of Radiocarbon Dating

In the "traditional" method of radiocarbon dating (prior to AMS), the age is determined by measuring the beta-decay rate of a carbon sample. The precision and the ultimate "limit" of the age measurement are governed by Poisson counting statistics and the signal-to-background ratio. Let S be the net sample count rate (cpm) and B be the detector background rate (cpm).

(a) A 1-gram "modern" carbon sample has a specific activity of $S_0 = 13.5$ cpm/g. A laboratory detector has a background rate of $B = 1.0$ cpm. If the sample is counted for $t = 1000$ minutes, the standard deviation of the net count rate for an old sample ($S \ll B$) is approximated by $\sigma \approx \sqrt{2B/t}$.

Calculate the age of a 1-gram sample at which the signal S is equal to twice the standard deviation ($S = 2\sigma$). Use $T_{1/2} = 5730$ years. This represents the practical dating limit (2σ detection limit).

(b) **Scaling Effects:** If you increase the sample size to 10 grams, the signal S_0 increases by a factor of 10. However, the background B also increases because a larger detector volume is required. Assume the background scales as $B \propto \text{Mass}^{0.7}$.

Calculate the new age limit ($S = 2\sigma$) for a 10-gram sample counted for the same 1000 minutes. Does the limit improve significantly?

(c) Explain why simply increasing the counting time t has a diminishing return on the maximum measurable age. Roughly how many times longer would you need to count to increase the age limit of the 1-gram sample by one additional half-life?

Problem 2: The N/Z Zig-Zag

A heavy nucleus like $^{238}_{92}\text{U}$ ($N/Z \approx 1.587$) is neutron-rich.

(a) Show mathematically that a single α-decay ($^{4}_{2}\text{He}$ emission) *increases* the N/Z ratio of the daughter nucleus. (Hint: Consider the inequality $\frac{N-2}{Z-2} > \frac{N}{Z}$ for $N > Z$).

(b) Calculate the exact N/Z ratio for the immediate daughter $^{234}_{90}\text{Th}$.

(c) Since the "Valley of Stability" curves such that lighter nuclei require a *lower* N/Z ratio, this alpha decay pushes the nucleus further into unstable (neutron-rich) territory. Which specific decay mode follows to "correct" this ratio back toward the stability line?

(d) How many total α and β^- decays are required to transform $^{238}_{92}\text{U}$ into stable $^{206}_{82}\text{Pb}$?

Problem 3: Radon Transport (The "Funnel" Effect)

A house with a basement footprint of $A_{slab} = 100\ \text{m}^2$ is built on a layer of highly permeable gravel. Beneath the gravel lies sandy soil with permeability $\kappa = 1.0 \times 10^{-11}\ \text{m}^2$ and gas viscosity $\mu = 1.8 \times 10^{-5}\ \text{Pa} \cdot \text{s}$.

Because the gravel layer allows gas to move freely, the "Stack Effect" suction from the house is distributed across the entire underside of the foundation. This creates a vertical pressure gradient of $\nabla P = 5\ \text{Pa/m}$ in the underlying soil, drawing gas up over the entire $100\ \text{m}^2$ footprint to eventually enter through perimeter cracks.

(a) Calculate the superficial vertical velocity v (m/s) of the soil gas through the sandy soil using Darcy's Law ($v = \frac{\kappa}{\mu}\nabla P$).

(b) Calculate the total volumetric flow rate Q (m^3/hr) of soil gas entering the gravel layer (and subsequently the basement).

(c) The soil gas has a Radon concentration of $[\text{Rn}]_{soil} = 2000$ pCi/L. The house volume is $500\ \text{m}^3$ with an air exchange rate of $\lambda_{air} = 0.5\ \text{hr}^{-1}$.

Find the steady-state indoor Radon concentration. Does this exceed the EPA Action Level of 4.0 pCi/L?

Problem 4: U-Pb Geochronology and the Concordia

A Zircon crystal yields the following molar ratios: $^{206}\text{Pb}/^{238}\text{U} = 0.60$ and $^{207}\text{Pb}/^{235}\text{U} = 42$.

(a) Calculate the age t using the ^{238}U clock ($\lambda_{238} = 1.551 \times 10^{-10}\ \text{yr}^{-1}$).

(b) Calculate the age t using the ^{235}U clock ($\lambda_{235} = 9.848 \times 10^{-10}\ \text{yr}^{-1}$).

(c) Based on your results, is this Zircon "Concordant" or "Discordant"? Explain physically what might cause these two clocks to disagree (i.e., did the crystal lose Uranium or Lead?).

Problem 5: The Earth's Nuclear Furnace

- The current total radiogenic heat release of the Earth is approximately 25 TW. If ^{238}U accounts for 40% of this power, calculate the total mass of ^{238}U currently in the Earth. (Specific power of pure $^{238}\text{U} = 9.4 \times 10^{-5}$ W/kg).
- Calculate the total radiogenic energy released over the age of the Earth (4.5×10^9 years) assuming the **average** radiogenic power over that time was 40 TW (higher in the past due to more isotopes).
- Compare this total energy to the estimated "Primordial Heat" of accretion (10^{31} J). Is nuclear decay a significant contributor to the Earth's thermal budget?